

Intermediate Macroeconomics: Lecture W6 (Ch.13)

ECON 311 – Intermediate Macroeconomics

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What are You Going to Learn (Ch. 13)

- The IS and MP curves combine to get the AD curve
- The Phillips curve can be shown as the AS curve
- Movements in the economy can be shown in the AS/AD model
- Modern monetary policy theories
- The appropriate policy response to shocks

Monetary Policy Rules and Aggregate Demand

Recall that:

- IS Curve:

$$\tilde{Y}_t = \bar{a} - \bar{b}(R_t - \bar{r})$$

- MP curve: Central bank pick R_t exogenously - so far
- Philips Curve:

$$\Delta\pi_t = \bar{\nu}\tilde{Y}_t + \bar{o}$$

- This model implies high short-run output ($\tilde{Y}_t$) increases inflation ($\Delta\pi_t > 0$)
- The central bank faces this trade-off and chooses the interest rate

Simple Monetary Policy Rule

- A monetary policy rule is a guideline for central bankers under certain economic circumstances, when output or inflation changes
- The following simple rule have the real interest rate a function of the inflation rate:
 - If inflation is high (low), the interest rate should increase (decrease)

$$R_t - \bar{r} = \bar{m}(\pi_t - \bar{\pi})$$

- The new component is
 - $\bar{\pi}$: Inflation target
 - As of now, Canada, the United States, United Kingdom, euro area, and Japan all have an explicit target of 2 percent
 - Treat it as the steady-state rate of inflation
 - $\bar{m}$: dictates how aggressively monetary policy responds to inflation

The AD Curve

- Put together the IS curve and the policy rule (sub in $R_t - \bar{r}$)

$$\tilde{Y} = \bar{a} - \bar{b}\bar{m}(\pi_t - \bar{\pi})$$

- Short-run output is a function of the inflation rate
- Invert the equation:

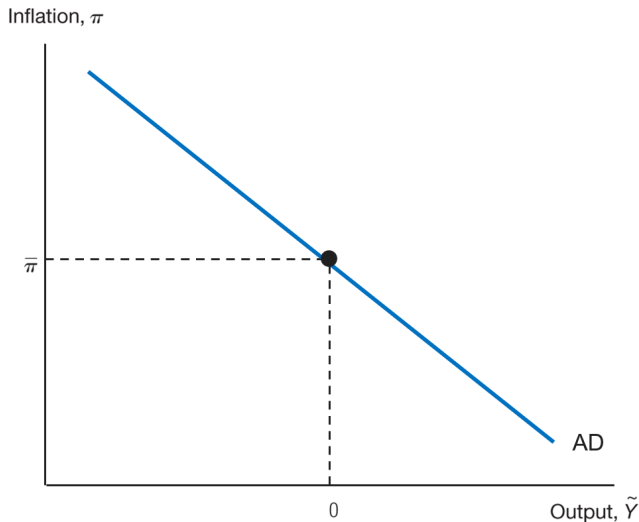
$$\tilde{Y} - \bar{a} = -\bar{b}\bar{m}(\pi_t - \bar{\pi})$$

$$\frac{\tilde{Y} - \bar{a}}{-\bar{b}\bar{m}} = (\pi_t - \bar{\pi})$$

$$\pi_t = \frac{\tilde{Y} - \bar{a}}{-\bar{b}\bar{m}} + \bar{\pi} = \frac{\bar{a}}{\bar{b}\bar{m}} + \bar{\pi} - \frac{1}{\bar{b}\bar{m}}\tilde{Y}$$

The AD Curve

- If inflation is above (below) its target, the Central Bank raises (falls) rates and output falls (rises) below potential



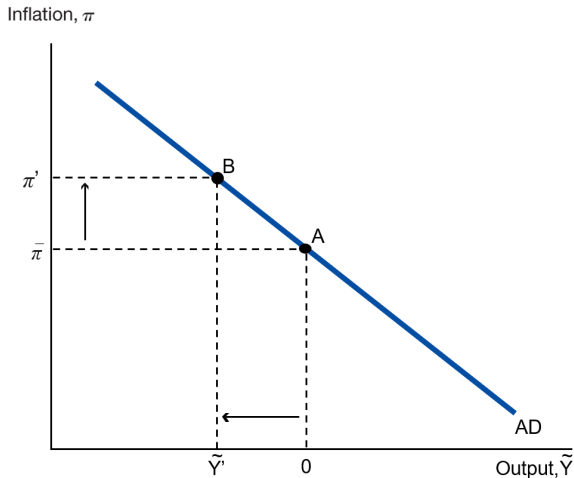
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The AD Curve

- The AD curve combine the central bank's choice of the interest rate, and, consequently, short-run output, based on the current inflation rate
 - Change in inflation: a movement along the AD curve
 - Change in $\bar{m}$: change the slope of the AD curve
 - Changes in the parameter $\bar{a}$ or $\bar{\pi}$: AD curve shifts

The AD Curve after an Inflation Shock

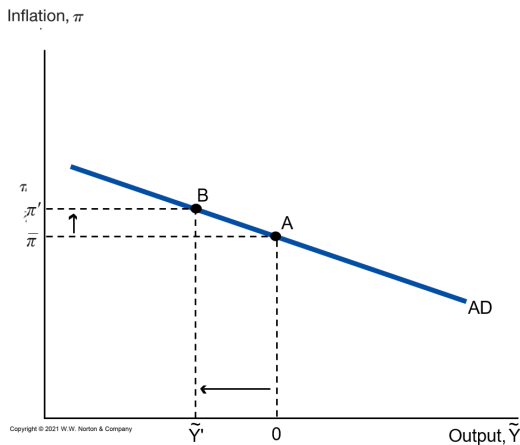
- Seeing an increase in inflation, the monetary policy rule dictates that the central bank increase the interest rate, thereby reducing short-run output



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An Aggressive Monetary Policy Rule

- A high value of $\bar{m}$ means the monetary policy rule produces a large recession in order to fight a given increase in the inflation rate
- A one-percentage-point increase in the inflation rate dictates a large decline in short-run output



The Aggregate Supply Curve

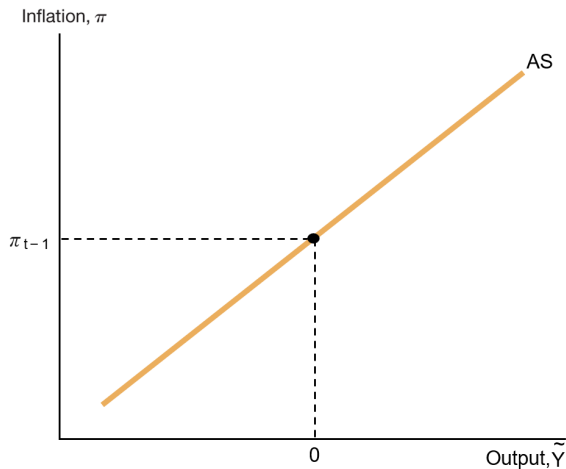
- The Phillips curve relates the current rate of inflation to short-run output
 - It describe the price-setting equation used by firms in the economy
 - Also, firms supply good and services

$$\pi_t = \pi_{t-1} + \bar{\nu} \tilde{Y}_t + \bar{o}$$

The Aggregate Supply Curve

- When there are no inflation shocks ($\bar{o} = 0$) and when output is at potential ($\tilde{Y} = 0$), firms simply expect the current rate of inflation to prevail (i.e., $\pi_t = \pi_{t-1}$)
- The AS curve will shift due to
 - change in $\tilde{Y}$
 - Higher output today leads firms to raise their prices by more, which increases inflation today and tomorrow
 - change in the inflation shock parameter
- These shock would carry over to future period until another shock cancel out the effect

The Aggregate Supply Curve



The AS/AD Framework

- Combining the AS and AD curve
- Two equation: AS and AD
- Two unknowns: inflation rate and short-run output

$$\tilde{Y} = \bar{a} - \bar{b}\bar{m}(\pi_t - \bar{\pi})$$

$$\pi_t = \pi_{t-1} + \bar{\nu}\tilde{Y}_t + \bar{o}$$

The Steady State

- We will first start with steady state, then study the dynamic of the model
- In the steady state
 - The endogenous variables are constant over time (i.e., $\pi_t = \pi_{t-1} = \pi^*$, $\tilde{Y} = 0$)
 - No shocks to the economy (i.e., $\bar{o} = 0, \bar{a} = 0$)
- Starting with the AS curve (let $\pi_t = \pi_{t-1} = \pi^*$):

$$\bar{\pi} = \bar{\pi} + \bar{\nu} \tilde{Y}_t \implies 0 = \bar{\nu} \tilde{Y}$$

$$\tilde{Y} = 0$$

- The AS curve is consistent with the steady state

The Steady State

- From the AD curve ($\tilde{Y} = 0$ and $\bar{a} = 0$)

$$\tilde{Y}_t = \bar{a} - \bar{b}\bar{m}(\pi_t - \bar{\pi})$$

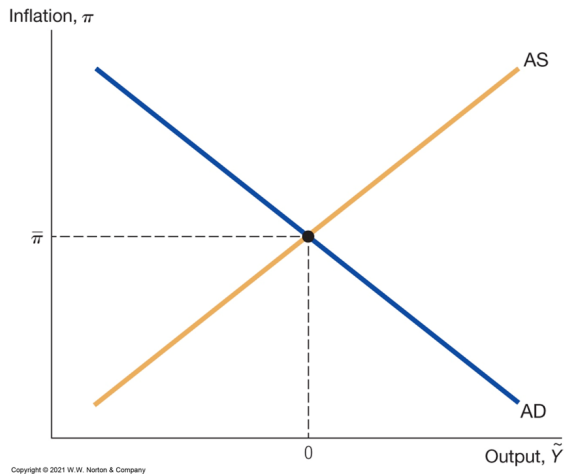
$$0 = 0 - \bar{b}\bar{m}(\pi_t - \bar{\pi})$$

- This implies

$$\pi_t = \pi_{t-1} = \bar{\pi}$$

- AD curve is also consistent with the steady state

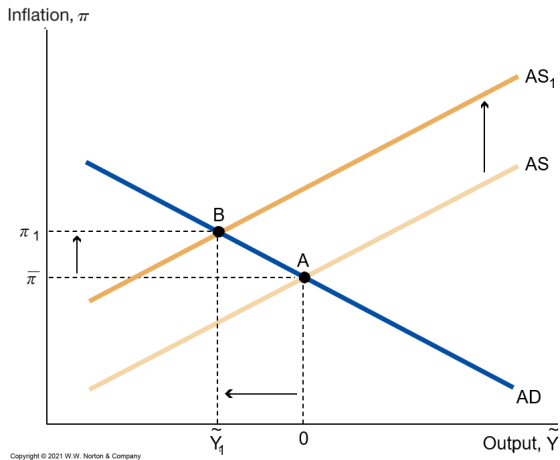
The AS/AD Framework



Macroeconomic Events: Event 1—An Inflation Shock

- Unless specific otherwise, the economy begins in steady state
- Suppose there is a increase oil price for one period
 - The parameter $\bar{o}$ is positive for one period
 - The inflation rises
 - The AS curve will shift up as a result
- The monetary policy rule states that the increase in inflation be fought by an increase in the real interest rate, reducing short-run output

The Initial Response to an Inflation Shock



- Stagflation: Combination of a recession and inflation
- But, that is just the initial impact

Event 1: An Inflation Shock

- In period 1, the inflation would be

$$\pi_1 = \bar{\pi} + \bar{\nu}\tilde{Y}_1 + \bar{o}$$

- In period 2, $\bar{o}$ return to zero, the AS curve is now:

$$\pi_2 = \pi_1 + \bar{\nu}\tilde{Y}_2 = \bar{\pi} + \bar{\nu}\tilde{Y}_1 + \bar{o} + \bar{\nu}\tilde{Y}_2$$

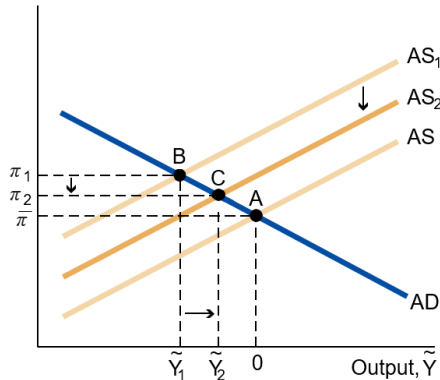
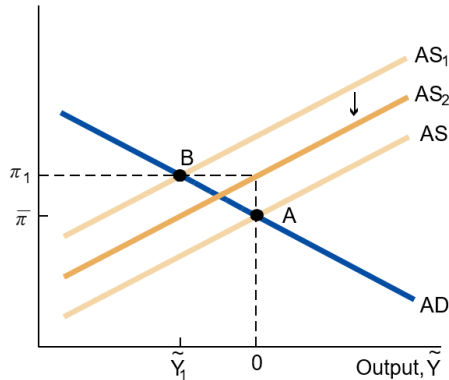
- Since $\pi_1 > \bar{\pi}$, the AS curve is not back to steady state
 - Note the term $\tilde{Y}_1$ and $\tilde{Y}_2$ are negative
 - This implies the inflation would get lower and lower over time!!

Event 1: An Inflation Shock

- In word,
 - High inflation from the oil shock
 - It rises the expected inflation
 - AS curve does not adjust back to steady state right away
 - As seen next, inflation slowly falls and model return to its steady state

Two Periods after an Inflation Shock

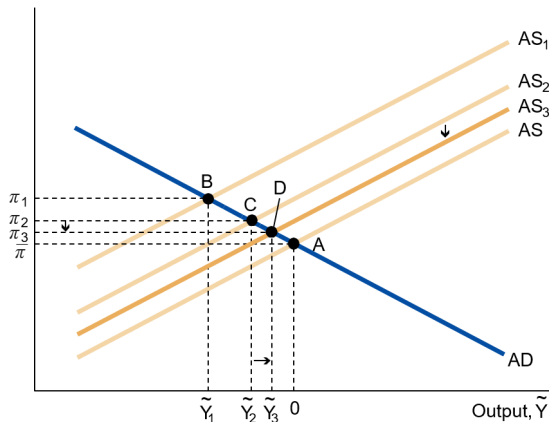
- Because of sticky prices, high inflation expectations adjust only slowly
- At $\tilde{Y} = 0$, the AS curve would shift down a bit from AS_1



Three Periods after an Inflation Shock

- In period three, expected inflation falls further, shifting the AS curve closer to its original position
- Eventually, the AS curve would move back to the steady state position

Three Periods after an Inflation Shock
Inflation, π

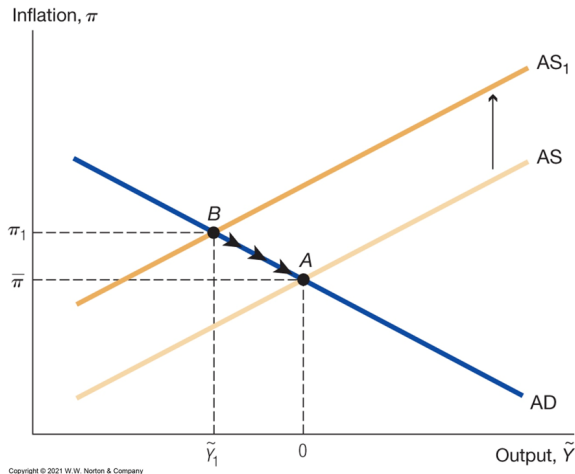


Event 1: An Inflation Shock

- What we have just saw is similar to the principle of transition dynamics, which we encountered in the long-run model
- Another note is that the movement back to the steady state is fastest when the economy is furthest from its steady state
- In summary, a price shock
 - raises inflation directly
 - the central bank induces a recession to fight inflation
 - keeps inflation higher for a longer period of time due to sticky inflation
 - results in a prolonged slump
 - causes the economy to suffer from stagflation

The Effects of an Inflation Shock

- As expected inflation declines gradually, causing the AS curve to shift back toward its original position
- Sticky inflation assumption is at the heart of this gradual adjustment



Event 2: Disinflation

- Suppose policymakers decide to lower the inflation target.
 - The AD curve shifts down, as $\bar{\pi}$ falls

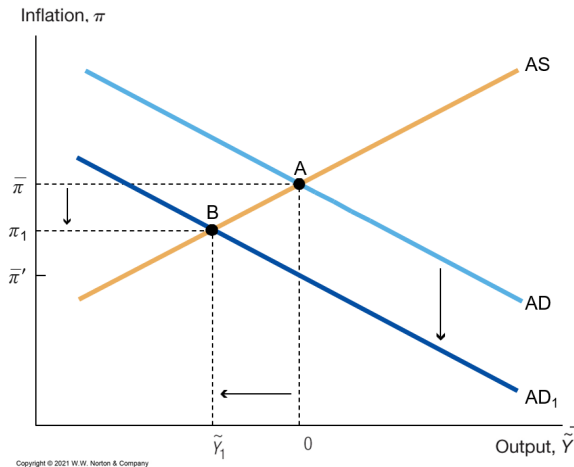
$$\tilde{Y}_t = \bar{a} - \bar{b}\bar{m}(\pi_t - \bar{\pi})$$

- The new rule calls for an increase in interest rates

$$R_t - \bar{r} = \bar{m}(\pi_t - \bar{\pi})$$

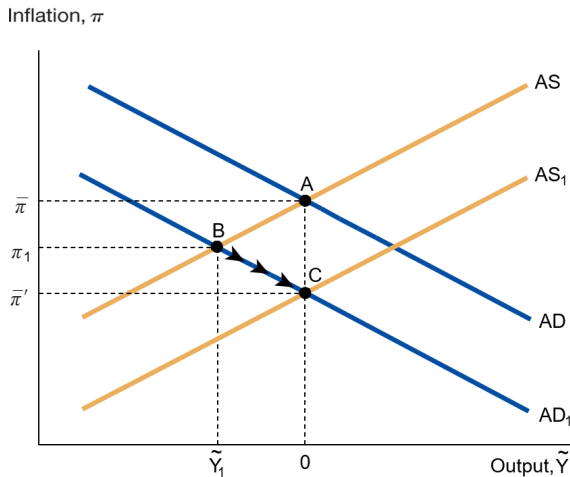
The Initial Response to Disinflation

- The reduction in the inflation target shifts the AD curve down



The Dynamics of Disinflation

- We can apply the principle of transition dynamics
- Here, the change in the monetary policy rule means that the steady-state rate of inflation has now changed



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Event 2: Disinflation

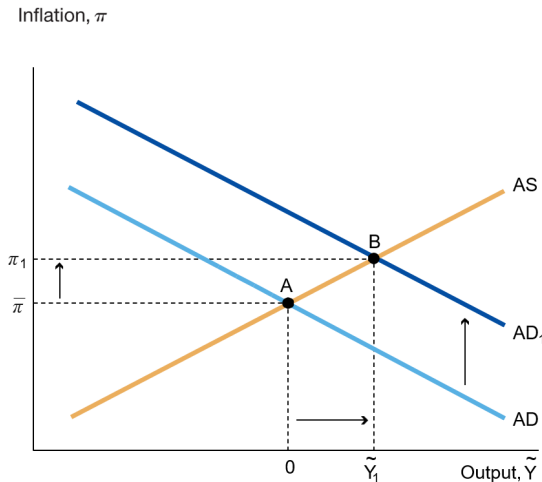
- If the classical dichotomy holds in the short run, the AD and AS curves would reach the new steady state immediately
- But there is a sticky inflation. A recession is needed to adjust expectations down
- In the process of changing a monetary policy rule, the credibility of policymakers and the way price setters form their expectations play critical roles
 - Michelle Alexopoulos (at U of T) and her co-authors analyze effect U.S. Federal Reserve Chairs's word, voice and facial expression when deliver congressional testimonies on financial markets (JME 2024)

Event 2: Disinflation

- Summary:
 - The policy change lead to a downward shift of the AD curve
 - The change in the rate of inflation causes the AS curve to shift in the following period
 - Firms adjust their expectation for inflation to account for the new lower inflation rate
 - The inflation rate is still above the target, the central bank keeps actual output below potential
 - Eventually, as inflation rate falls further, the economy will rest in its new steady state

Event 3: A Positive AD Shock

- Suppose there is a temporary increase in the aggregate demand parameter, $\bar{a}$
- The AD curve will shift out
 - Increased demand lead to an increase in prices

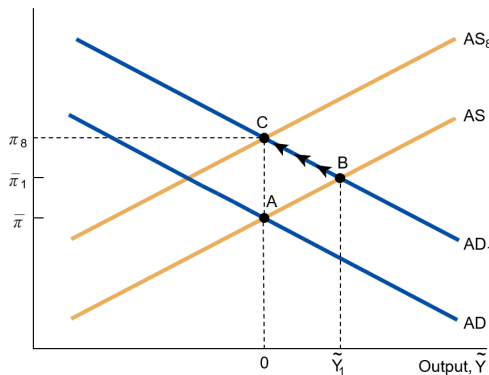


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Event 3: A Positive AD Shock

- Following the increase in demand (i.e., $\bar{a}$ increases), firms raise prices.
 - Inflation increases
- As the inflation rise, AS curve shift up
- Short-run output return to zero gradually

Dynamics as the AS Curve Shifts
Inflation, π

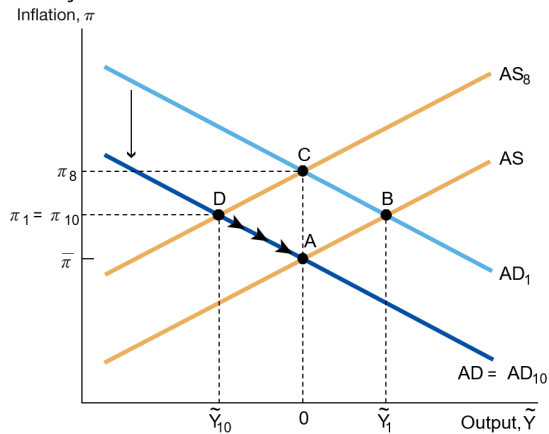


The Unraveling after the AD Shock Ends

- Recall, aggregate demand shocks are temporary. In the long run, $\bar{a} = 0$
- When the shock ends, the AD curve shifts back to its original position
 - This is much like a negative shock because short-run output was already at 0
- The shift back to its original position puts downward pressure on inflation:
 - Transition dynamics returns the economy to steady state as the AS curve shifts

The Unraveling after the AD Shock Ends

The Unraveling after the AD Shock Ends



Event 3: A Positive AD Shock

- The AD shock implies that booms are matched by recessions
 - The economy benefits from a boom but inflation rises
 - The way to reduce inflation is by a recession
- The costs of inflation: The economy would have been better off at the original steady state

Empirical Evidence: Predicting the Central Bank Policy Rate

- Does simple monetary policy rule provides an accurate picture of the federal funds rate?
- Recall the policy rule and the Fisher's equation

$$R_t - \bar{r} = \bar{m}(\pi_t - \bar{\pi})$$

$$i_t = R_t + \pi_t$$

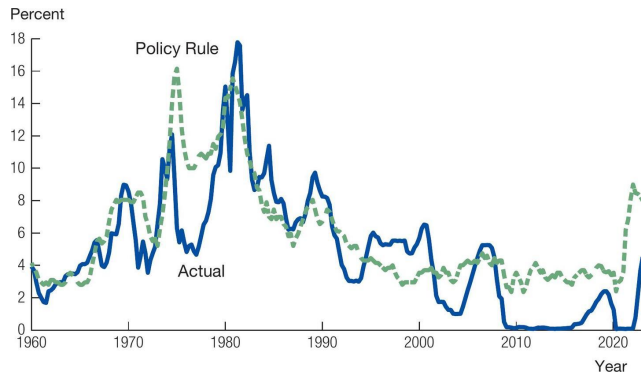
- Use the Fisher equation to write the monetary policy rule in terms of the nominal interest rate:

$$i_t = \bar{r} + \pi_t + \bar{m}(\pi_t - \bar{\pi})$$

- It is commonly known as the Taylor's rule
 - The Taylor's rule typically also have output build into it

The Fed Funds Rate, Actual and Predicted

- Predicted level for the fed funds rate using the simple policy rule
 - $\bar{m} = 1/2, \bar{r} = 2\%, \bar{\pi} = 2\%$
- Some economists infer that monetary policy was excessively loose during 70s, resulting the high inflation in the 70s
- The Fed also departs from the rule during recessions in the early 1990s, in the early 2000s, and since 2008



Next Lecture

- Next Lecture will cover Ch.18